[Corresponding author name]
School of Electrical and Computer Engineering
College of Engineering, University of Tehran
Tehran, Iran
[email]

October 16, 2026

The Editors
npj Digital Medicine

Dear Editors,

We submit the Article "Separating decision rule failures from representation failures in external validation of diabetic retinopathy grading" for consideration in *npj Digital Medicine*.

When a diabetic retinopathy grading model fails on a new population, the field attributes the failure to the learned representation and responds by changing the model: a larger backbone, a foundation encoder, a new ordinal loss. We separate that contribution from the decision rule that turns a model's ordinal outputs into a grade. Across ten development models, two external populations (APTOS 2019 and DDR), five training seeds and the reverse direction of transfer, the ranking of images by severity transferred while the mapping from score to grade did not: the same models over-graded one population and under-graded the other, so a referral threshold fitted for 85% sensitivity reached 99.33% and 63.38%. Refitting four cut-points alone restored the mild grade on APTOS and raised referral sensitivity in the reverse direction from 43.5% to 77.9%; where the ranking itself failed, recalibration could not help. We report with equal weight that two of our own pre-registered predictions were falsified: the mild-grade collapse did not generalise beyond APTOS, and models differed more than deployable rules. Reanalysing the released predictions of a published ordinal model reproduces its result and shows the same rule dependence, and 8 of 9 pre-registered interventions on the representation, including a retinal foundation model, were falsified at matched calibration.

We think the work suits *npj Digital Medicine* for three reasons. First, its message is practical for deployment: before a grading model is replaced, its decision rule should be recalibrated and evaluated against its full operating curve, and we quantify how many locally labelled images that requires and how often it does harm. Second, it corrects how model comparisons are reported: we propose *matched calibration* as a reporting rule and document its record, including a case in which our own earlier archived result did not survive regeneration from per-image predictions. Third, the study is fully reproducible: every outcome was pre-registered in a public repository before the corresponding run, every number in the manuscript is generated from archived per-image predictions, and code, splits and predictions are public.

The manuscript has not been published and is not under consideration elsewhere. All data are public and de-identified. The authors declare no competing interests. The use of an AI assistant in analysis and drafting is disclosed in Methods.

[Optional: suggested and excluded reviewers, to be added by the authors.]

Sincerely,

[Corresponding author name]
School of Electrical and Computer Engineering
University of Tehran